

Mohanbalaji Devarapu | Data Analyst

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SUMMARY

Data Analyst & Analytics Engineer with 3+ years of experience in healthcare and IT, specializing in data modeling, ETL pipelines, predictive analytics, and BI dashboards. Skilled in Python, R, SQL, dbt, Airflow, Snowflake, Power BI, Tableau, and BigQuery to turn complex datasets into actionable insights. Experienced in healthcare analytics, KPI reporting, ad-hoc analysis, and data transformation, delivering measurable improvements in operational efficiency and cost reduction. Adept at collaborating with cross-functional teams and ensuring regulatory compliance (HIPAA, GDPR, CMS, HL7, FHIR).

TECHNICAL SKILLS

Languages & Scripting:	Python, R, SQL (Advanced – Joins, CTEs, Window Functions, Query Optimization, Stored Procedures), VBA, Power Query (M), Bash
Frameworks & Libraries:	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly, statsmodels, SciPy, spaCy, NLTK, dbt
Databases & Warehousing:	SQL Server, Snowflake, Amazon Redshift, Google BigQuery, Data Modeling (ER Diagrams), Metadata Management, Data Lineage, Validation Frameworks
Visualization & Business Intelligence:	Power BI (DAX, Power Query, Funnel Charts, KPI Reporting, Ad-hoc Analysis), Tableau (LOD Calculations, Parameter Actions, Trend Analysis), Qlik, Excel (Pivot Tables, Advanced Formulas, Scenarios, VBA), Looker Studio, Canva
Analytics & Statistical Skills:	Exploratory Data Analysis (EDA), Predictive Modeling, Feature Engineering, Statistical Analyses, Regression Analysis, Hypothesis Testing, A/B Testing, Cohort Analysis, Text Classification, Named Entity Recognition
Data Pipelines & Analytics Engineering:	Apache Airflow, PySpark, Python-based ETL Pipelines, dbt, ETL/ELT Pipelines, Data Cleaning, Data Transformation
Project Management & Collaboration	Agile/Scrum, JIRA, Confluence, CRM/ERP Platforms, Consulting Environment Exposure, Stakeholder Collaboration
Regulatory & Compliance:	HIPAA, GDPR, CCPA, CMS Star Ratings, HEDIS, HL7, FHIR, Data Anonymization, Audit Compliance, Bias Detection
Business & Operational Impact:	Business Transformation, Operational Efficiency, Digital Innovation

EXPERIENCE

Data Analyst | Humana Inc., KS

Jun 2025 – Present

- Consolidated medical claims, pharmacy, and EHR datasets into Snowflake, documenting data dictionaries and lineage, reducing metric inconsistencies across business units by 28%.
- Partnered with clinicians, actuaries, and operations leaders to define utilization KPIs (ER visits, readmission rates, drug adherence), mapping each KPI to specific claims and provider feeds.
- Automated ETL workflows with dbt and Apache Airflow, transforming raw claim transactions into member-level cost & utilization fact tables, enabling hourly refresh for real-time analytics.
- Built predictive cost-risk models in Python (Scikit-learn, SHAP) to identify members likely to incur avoidable hospitalizations, reducing preventable admissions by 11%.
- Performed quality audits in SQL Server to validate claims vs EHR encounters, detecting and correcting mismatches worth \$2.3M in reporting errors.
- Designed Power BI dashboards visualizing ER utilization, pharmacy spend, and readmission trends, supporting executive interventions that lowered total medical cost by 7% YoY.

Data Analyst Intern | Humana Inc., KS

Mar 2025 – May 2025

- Documented source-to-target mappings in Confluence for claims, pharmacy, and EHR feeds, supporting downstream analytics and ensuring traceability.
- Profiled raw claims datasets using Excel Power Query, identifying duplicates and incomplete provider IDs that caused 12% delays in claims adjudication.
- Assisted in staging claims data pipelines by developing SQL scripts for initial extraction and validation, ensuring completeness before loading into Snowflake.
- Conducted exploratory analysis in R (ggplot2, dplyr) to uncover utilization patterns across demographics, highlighting higher ER dependency in Tier-3 regions.
- Supported pilot testing of readmission risk models by validating training/test data splits in BigQuery, ensuring statistical balance across cohorts.
- Built Tableau reports summarizing hospital utilization and cost trends for monthly stakeholder reviews.

Data Analyst | Adons Softech, India

Jun 2022 – Jun 2023

- Standardized claims lifecycle data models using dbt and SQL, enforcing consistent metric definitions across underwriting, operations, and finance teams, reducing reporting discrepancies by 30%.
- Analyzed claim rejection trends using Python (Pandas, Seaborn) for healthcare processes, improving first-pass claim approvals by 17%.

- Audited premium payments and claim disbursement logs with SQL Server and Excel reconciliation routines, identifying mismatches worth ₹1.2M and accelerating financial closure.
- Explored policy lapse patterns by region/demographics using R (ggplot2, dplyr), improving renewal rates in Tier-2 regions by 12%.
- Built Power BI dashboards tracking turnaround time (TAT) across claim categories, shortening processing cycles by 20%.
- Delivered executive-level insights via Looker Studio and Canva, maintaining compliance datasets with SharePoint audit trails.

Data Analyst | Tophat Software Technologies, India

Jan 2021 – May 2022

- Standardized event taxonomy and KPI definitions, creating a centralized data dictionary, improving reporting accuracy by 35%.
- Aligned 10+ product/marketing objectives with reliable data sources, reducing ad-hoc reporting requests by 25%.
- Engineered real-time ETL pipelines using Kafka, AWS S3, and Snowflake, achieving <60s data freshness.
- Performed cohort, funnel, and anomaly analyses with SQL/Python, increasing user retention by 12% and reducing missed defect detection by 30%.
- Designed and validated A/B experiments, using statistical testing to recommend campaign/feature rollouts, boosting conversion rates by 15%.
- Built interactive Power BI dashboards with automated refresh and alerting, reducing reporting turnaround time by **40%** and improving incident response by 20%.

EDUCATION

Master of Science in Information Studies	May 2025
Trine University, Indiana	
Bachelor in Computer Science & Engineering	Jun 2022
QIS College of Engineering and Technology, India	

PROJECTS

SAP Ariba Buying & Procurement Process Optimization

- Streamlined procurement workflows using SQL, Excel, and ServiceNow, improving purchase order processing efficiency by 15%.
- Coordinated vendor engagement, goods delivery, and installation while tracking operational KPIs in Excel dashboards, enhancing procurement transparency.
- Implemented process improvements and automated reporting, supporting data-driven decision-making and aligning workflows with organizational standards.

Technical Support and Service Quality Enhancement for Wells Fargo Advisors

- Analyzed service request logs using SQL and Excel Power Query, identifying recurring bottlenecks and reducing issue resolution time by 20%.
- Built dashboards in Power BI to monitor issue trends, resolution performance, and service quality compliance.
- Collaborated with cross-functional teams to implement process improvements, ensuring alignment with internal standards and enhancing operational efficiency.